MIL-C-24301B(EC)
3 October 1969
SUPERSEDING
MIL-C-24301A(EC)
14 March 1969

MILITARY SPECIFICATION

CABLE, SPECIAL PURPOSE, RADIO FREQUENCY, BUOYANT, TYPE RG-374/U

1. SCOPE

1.1 This specification covers special single-conductor buoyant cable with high mechanical performance characteristics, such as high strength, flexibility, low compression set, low water absorption, low temperature properties, and good abrasion resistance.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-390	—	Plastic Molding Material, Polyethylene, Low and Medium Density.
QQ-W-343	—	Wire, Electrical (Uninsulated).

MILITARY

MIL-C-17	—	Cables, Radio Frequency; Coaxial, Dual Coaxial, Twin Conductor, and Twin Lead.
MIL-Q-9858	—	Quality Program Requirements.
# MIL-C-12000	—	Cable, Cord, and Wire, Electric; Packaging of.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article sample. - Prior to beginning production a sample shall be tested as specified in 4.3.1 (see 6.3).

3.2 Materials and construction. -

3.2.1 Center conductor. - The center conductor of the cable shall be #18 AWG, Type S drawn and fully annealed, bare copper wire in accordance with QQ-W-343 with a maximum direct current (d.c.) resistance of 6.4 ohms per 1000 feet at 20°C., and a continuous length with no joints or splices.

3.2.2 Solid core dielectric. - The solid core shall be polyethylene in accordance with type II, class L, grade 7 (natural) or 7a (colors) of L-P-390 having an outside diameter of 0.160 ±0.004 inch.

3.2.3 Strength members. - The strength members shall consist of 11 strands of 0.050-inch diameter 1100 Denier prestretched polyester fibre with a 20-inch left hand lay. Each strand shall be made up of 1100 Denier ±3 percent prestretched polyester yarn, multi-filament 0 to ½ turns per inch, bright high tenacity, 16 ends, 0.9 ±10 percent turns per inch, strength 220 pounds minimum elongation 3 percent maximum at 20 percent ultimate and 8.5 percent maximum at ultimate.

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3.2.4 Jacket. - The jacket shall consist of two layers of continuous, homogeneous unicellular black foamed polyethylene. The extreme O.D. of the cable shall be 0.650 ±0.025 inch. The outer surface of the cable shall be smooth so that an O-ring of 0.360 inch cross section and 0.620 inch I.D. (Federal Stock #H-5330-064-6585) will seal anywhere along the entire length of the cable without leakage at any pressure from 0 - 600 pounds per square inch gage (psig) when applied for 2 hours. The outer surface shall be of uniform hardness and free of major imperfections such as blow holes, cuts, valleys and bruises and abrupt changes of diameter within the tolerance range.

3.3 Breaking strength. - The cable shall have a minimum breaking strength of 1400 pounds when tested in accordance with 4.8.1.

3.4 Shear strength. -

3.4.1 Between center conductor and solid core dielectric. - The shear strength between the center conductor and the solid core dielectric shall be a minimum of 20 pounds for each linear foot of cable.

3.4.2 Between dacron and jacket. - The shear strength between the dacron and the jacket shall be 100 pounds minimum for each linear foot of cable. The shear strength between each layer of cable jacket shall be 100 pounds for each linear foot.

3.5 Specific gravity. - The cable shall have the lowest possible overall specific gravity consistent with the requirements specified herein. The specific gravity shall not exceed 0.82 when tested as specified in 4.8.3.

3.6 Color bend. - When tested as specified in 4.8.4 there shall be no evidence of cracks or flaws in either the dielectric material or the jacket.

3.7 Crack resistance. - The cable shall be resistant to stress cracking when tested as specified in 4.8.5.

3.8 Length. - Unless otherwise specified in the contract or order, the cable shall be provided in not less than 2000-foot nor more than 2100-foot continuous lengths (with no joints or splices).

3.9 Marking. -

3.9.1 Manufacturer's identification. - Manufacturer's identification shall consist of coloring the solid dielectric core in accordance with a manufacturer's identification color to be assigned by the command or agency concerned. Coloring of the core shall not affect the characteristics of the cable. There shall be no markings on the outside surface of the cable.

3.9.2 Reel marking. - All reels shall be marked with cable designation, contract number, date of manufacture, manufacturer, and length of cable.

3.10 Workmanship. - The workmanship shall be first class in every respect and uniform through each lot of cable offered for delivery.

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4. QUALITY ASSURANCE PROVISIONS

4.1 *Responsibility for inspection.* - *Unless otherwise specified* in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. - The inspection requirements specified herein are classified as follows:

(a) First article inspection (4.3)
(b) Quality conformance inspection (4.4)

4.3 First article inspection. - First article inspection shall consist of the examination and tests specified in 4.6 and 4.8 to determine conformance with this specification.

4.3.1 First article sample. - A 300-foot sample of cable constructed with specified materials and manufactured with equipment to be used in production shall be subjected to the first article inspection specified in 4.3. Unless otherwise instructed, the first article inspection shall be conducted at the Naval Ship Engineering Center, Philadelphia Division, Submarine Antenna Quality Assurance Department (SAQAD), Code 6700D. The above sample shall be furnished to Code 6700D with complete test data of 4.6. A test report shall then be forwarded by SAQAD to the contracting officer and contractor with an information copy to the Naval Electronic Systems Command (05143). The manufacturer shall not initiate production prior to receipt of approval by the contracting officer except at his own risk.

4.4 Quality conformance inspection. - The quality conformance inspection shall consist of production inspection (4.5).

4.5 Production inspection. - The production inspection shall consist of visual and dimensional examinations of 4.6 conducted by the manufacturer and breaking strength, shear strength, and specific gravity tests of 4.8.1, 4.8.2 and 4.8.3 conducted by SAQAD.

4.5.1 Sampling for production inspection. - Each completed length of cable shall be subjected to visual and dimensional examinations in accordance with 4.6. A 10-foot specimen from each reel of cable shall be forwarded to SAQAD. The data on the visual and dimensional examinations on each completed length of cable shall accompany the specimens forwarded to SAQAD.

4.5.2 Acceptance or rejection. - Acceptance or rejection of the cable will be based on an evaluation of the visual and dimensional examination data and the result of the breaking strength, shear strength and specific gravity tests. Acceptance or rejection of the cable will be provided by SAQAD within 14 days after receipt of specimens. Cable shall not be shipped until receipt of acceptance from SAQAD.

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4.6 Visual and dimensional examination. - The cable shall be examined to verify that the design, construction, physical dimensions, and workmanship are in accordance with this specification. The manufacturer shall provide a monitoring system whereby the overall diameters of the dielectric, and outer jacket are to be measured. The cable making equipment shall be fitted with electronic or mechanical measuring device(s) to produce continuous graphic records of the outside diameter of the dielectric and outer jacket of the cable. Two recordings are to be made as nearly simultaneously as possible of each diameter, 90 degrees apart, and at a point in manufacturing where further dimensional changes will not occur, such as those resulting from cooling, tension, reeling, etc. The records are to be permanent and reproducible by a common commercial process. Recording accuracy shall be no less than plus or minus 5 percent of the tolerance being measured and any portion of the recording of the jacket diameter shall be identified to within plus or minus 2 feet with the cable to which it applies. The records are to be used by Government personnel for rapid evaluation and in the quality control of the cable and shall be forwarded with the 10-foot length required by 4.5.1.

4.7 Quality control system. - The contractor shall establish and maintain an effective quality control system in accordance with MIL-Q-9858.

4.8 Test procedures. -

4.8.1 Cable strength. - Minimum tensile strength of the cable shall be 1400 pounds. The cable strength shall be determined by means of a power driven tensile machine. The rate of travel of the power actuated grip shall be adjusted to move at a rate of 12 ±2 feet every minute. Distance between grips shall be 5 feet prior to the application of the load. A reduction of the jacket diameter to 0.580 inch shall be considered a failure.

4.8.2 Shear strength. - A suitable device shall be constructed to measure the shear strength between each bond surface. 1-foot pieces shall be used for the tests. Different samples may be used to measure the bond between surfaces. A suitable device such as a Kellems grip or equivalent shall be used to facilitate measurement of the shear strength between each bonded surface. A satisfactory test between the strength members and outer jacket shall be an indication that the shear strength between the various layers is satisfactory.

4.8.3 Specific gravity. - The specific gravity shall not exceed 0.82 when measured over a hydrostatic pressure of 600 psig at room temperature in fresh water. The measurement shall be made after a continuous immersion time of 2 hours at 600 psig. The cable sample shall not be removed from the pressure tank prior to completion of required measurements.

4.8.4 Cold bend. - The cable shall be subjected to the cold bend test of MIL-C-17.

4.8.5 Crack resistance. - The cable shall be resistant to stress cracking when tightly wrapped around a mandrel 3 inches in diameter for a continuous 24-hour period of time.

4.9 Inspection conditions. - Unless otherwise specified herein, all inspection shall be made at room ambient temperature, pressure, and humidity.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

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5.1 Preservation, packaging, packing and marking. - Unless otherwise specified in the contract or order preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing and marking specified in MIL-C-12000 (see 6.1).

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of preservation, packaging, packing and marking required (see 5.1).

6.2 Consulting services. - Technical consulting services of the Naval Ship Engineering Center, Philadelphia Division, Submarine Antenna Quality Assurance Department, Code 6700D will be available until satisfactory completion of the quality conformance inspection.

6.3 First article. -

6.3.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement (see 4.3.1).

6.4 Changes from previous issue. - The outside margins of this document have been marked "#" to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
NAVY - EC

(Project 6145-N103)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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